Ezra Mohammed

Hardware Design Engineer

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easiyo

Summary

Hardware Design Engineer specializing in PCB layout, analog/digital circuit design, and standards-compliant electronics for IoT and smart metering systems. Expert in Altium Designer and EasyEDA, delivering production-ready, low-power hardware for national infrastructure projects.

Professional Experience

Jul 2025 – Present **Project Manager**, *DAFTech Social ICT Solution PLC*, Addis Ababa, Ethiopia, Skills – Altium Designer, EasyEDA, PCB Layout, ISO/IEC Compliance

- Led hardware design strategy for three national IoT/IIoT projects, including Ethiopia's first smart energy and water metering systems.
- Oversaw full product lifecycle from schematic to mass production, ensuring ISO/IEC compliance and regulatory approval.
- Managed PCB design, component selection, and layout optimization under budget and timeline constraints.

Apr 2024 – Jul 2025 **Hardware Design Engineer**, *DAFTech Social ICT Solution PLC*, Addis Ababa, Ethiopia, Skills – Altium Designer, EasyEDA, PCB Design, Wireless Integration

- Designed complete PCB hardware for Ethiopia's first prepaid smart energy meter, including multi-layer layouts and RF modules.
- Engineered hardware platforms for electromagnetic AMR water meter readers and ultrasonic smart water meters.
- Managed end-to-end PCB design using Altium Designer and EasyEDA, integrating microcontrollers, sensors, and wireless protocols.
- Optimized PCB layouts for low-power operation, signal integrity, and manufacturability.

Sep 2024 – Sep 2025 **Hardware Design Engineer**, *Ciket Technology*, Addis Ababa, Ethiopia, Skills – Altium Designer, PCB Design, Linux

- Designed and implemented full hardware platform for a Smart Fleet Management System.
- Engineered complete PCB layout and design process using Altium Designer, ensuring reliability in harsh environments.

Projects

Smart Prepaid Single-Phase & Three-Phase Watt-Hour Meter | *Altium Designer, Multi-Protocol RF, ISO/IEC Standards*

- Co-designed PCBs using Altium Designer, compliant with ISO/IEC standards (under final testing).
- Engineered multi-layer boards supporting NB-IoT, Cat-M1, Zigbee, LoRaWAN, and GSM.
- Designed hardware to support 100% digital transactions via e-banking and USSD.

LoRa/LoRaWAN Powered Smart Water Meter | *Altium Designer, Ultrasonic Sensors, Low-Power Design*

- Co-designed hardware for ultrasonic flow measurement with 98% accuracy.
- Implemented low-power LoRa/LoRaWAN communication for real-time remote monitoring.
- Led transition from R&D to mass production: 5,000–10,000 units delivered to AAWSA.

Digital Water Meter Reader (DMR) – Analog Meters | *Altium Designer, LC Sensors, GSM*

- Engineered cost-effective LC sensor-based digitization system to extend legacy meter lifespan.
- Designed compact PCB hardware with GSM connectivity for remote monitoring.
- Deployed 1,000 units across Addis Ababa and Adama.

Technical Skills

Electronics Design	Altium Designer, EasyEDA, KiCad, Proteus, Quartus, DipTrace, SCADA
Circuit & EM Simulation	ModelSIM, Ansys HFSS, LABView
Tools & Platforms	GitHub, Linux, SQL/HeidiSQL

Education

2021 – 2025 **B.Sc. in Microelectronics**, *School of Electrical and Computer Engineering, Addis Ababa Institute of Technology (AAiT)*, Addis Ababa University, Addis Ababa, Ethiopia

Certifications
Advanced PCB Design Certification (Altium Academy)
Advanced Embedded Systems (Udemy)